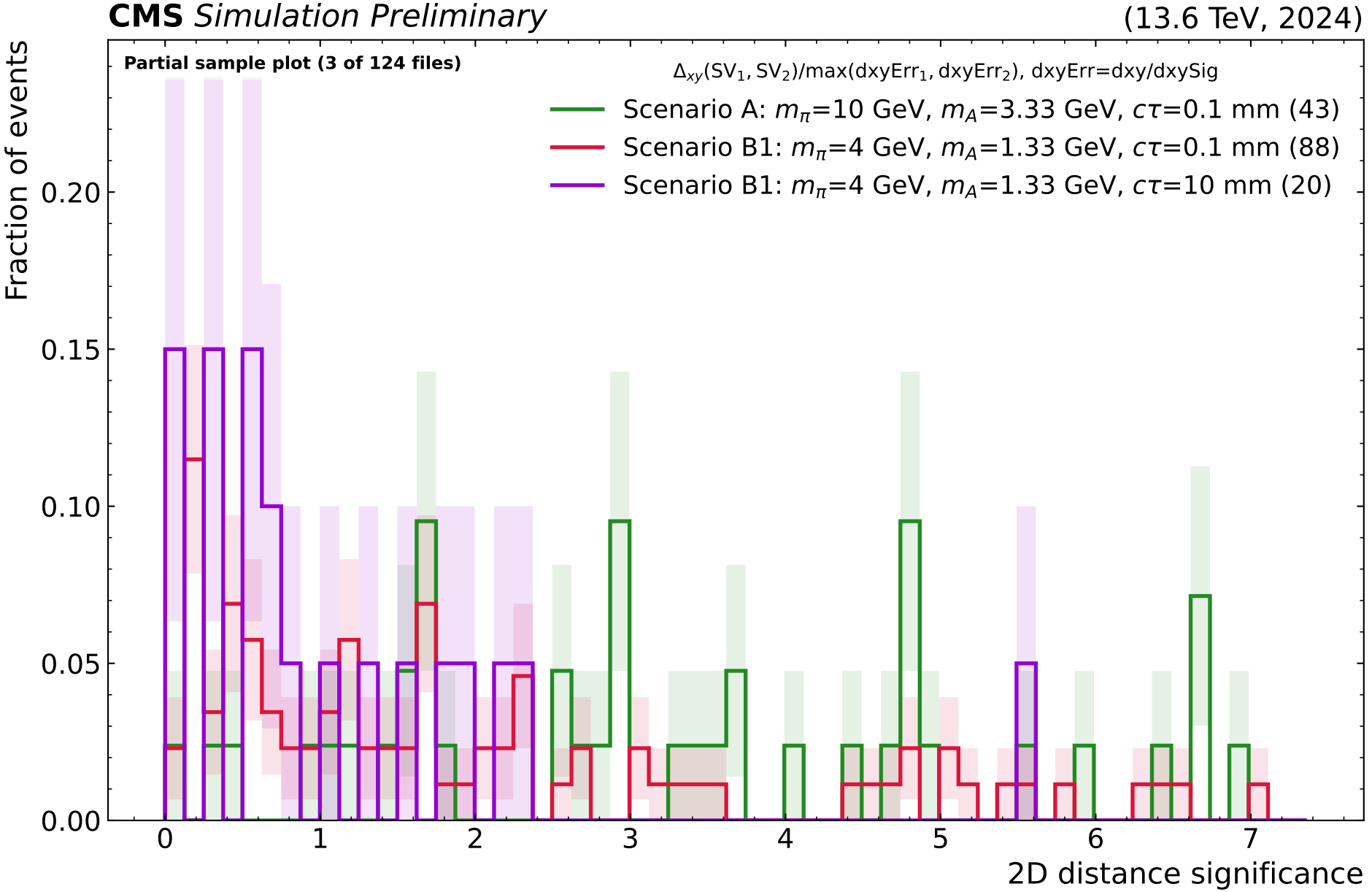
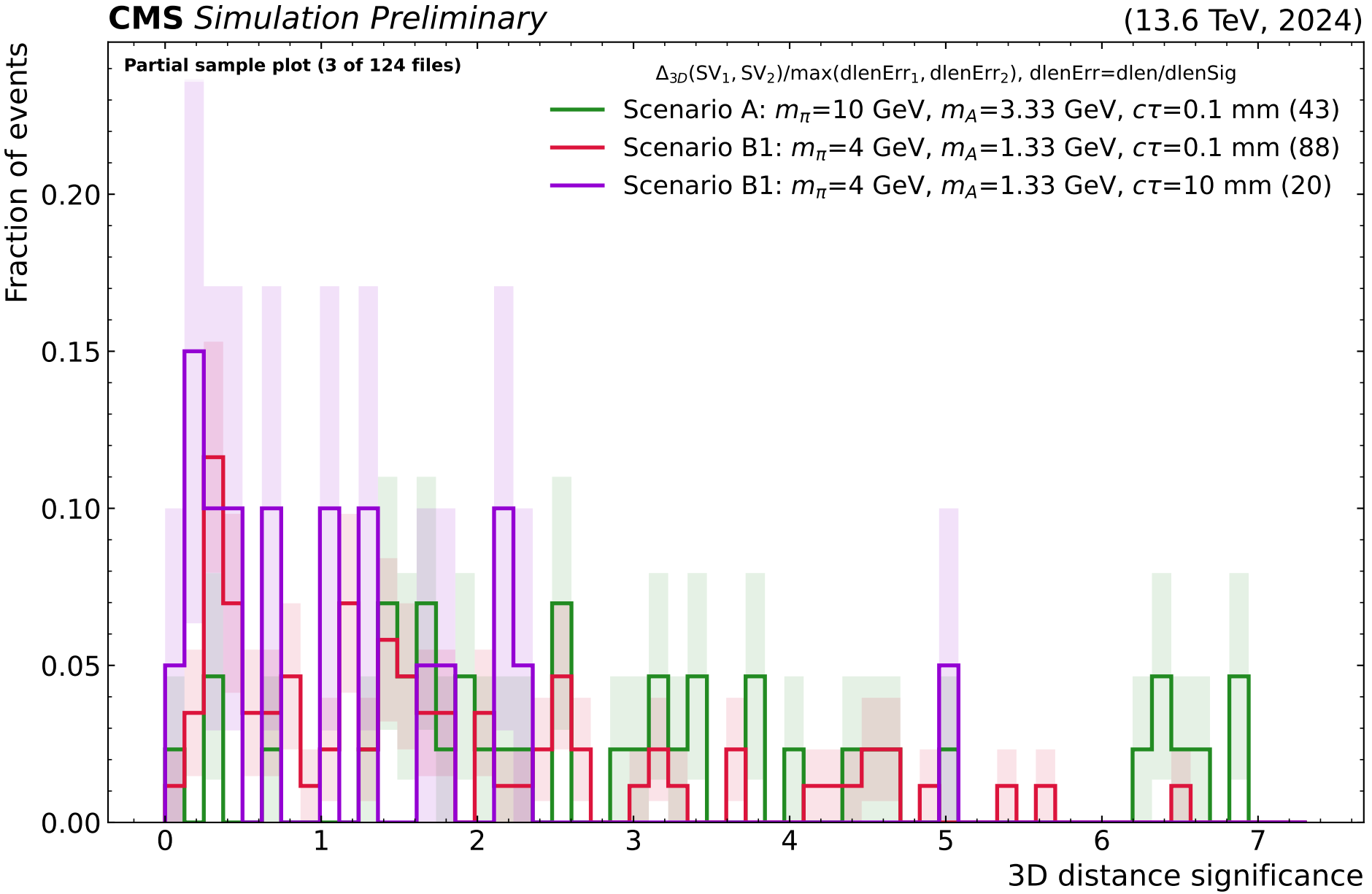
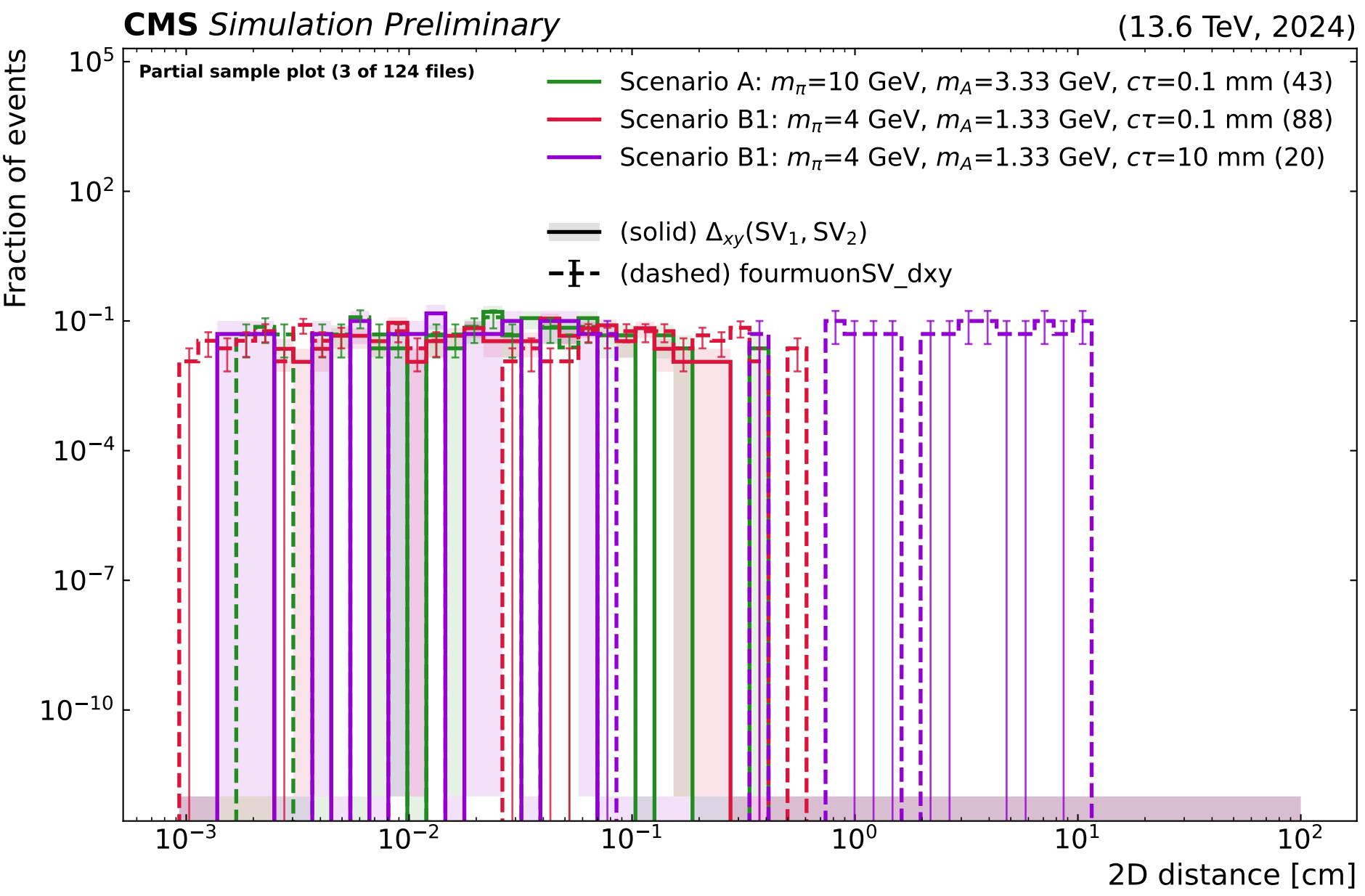
quadv: fourmuonSV + 2 consistent muonSV
trigger (Mu10 OR DoubleMu) + offline kinematic pre-selection
each distribution is normalised to unit area; bands/bars are statistical only



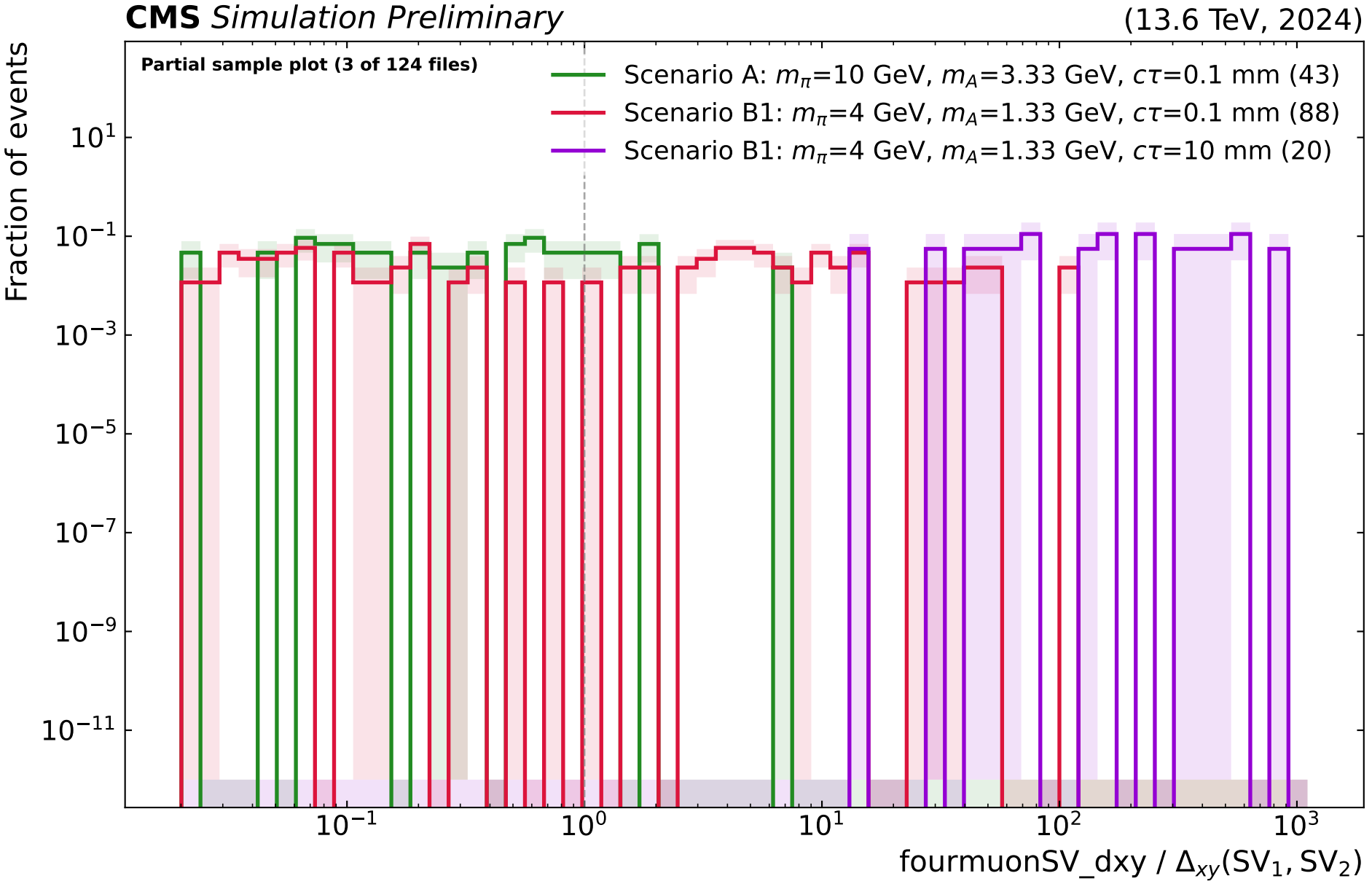
quadv: fourmuonSV + 2 consistent muonSV
trigger (Mu10 OR DoubleMu) + offline kinematic pre-selection
each distribution is normalised to unit area; bands/bars are statistical only



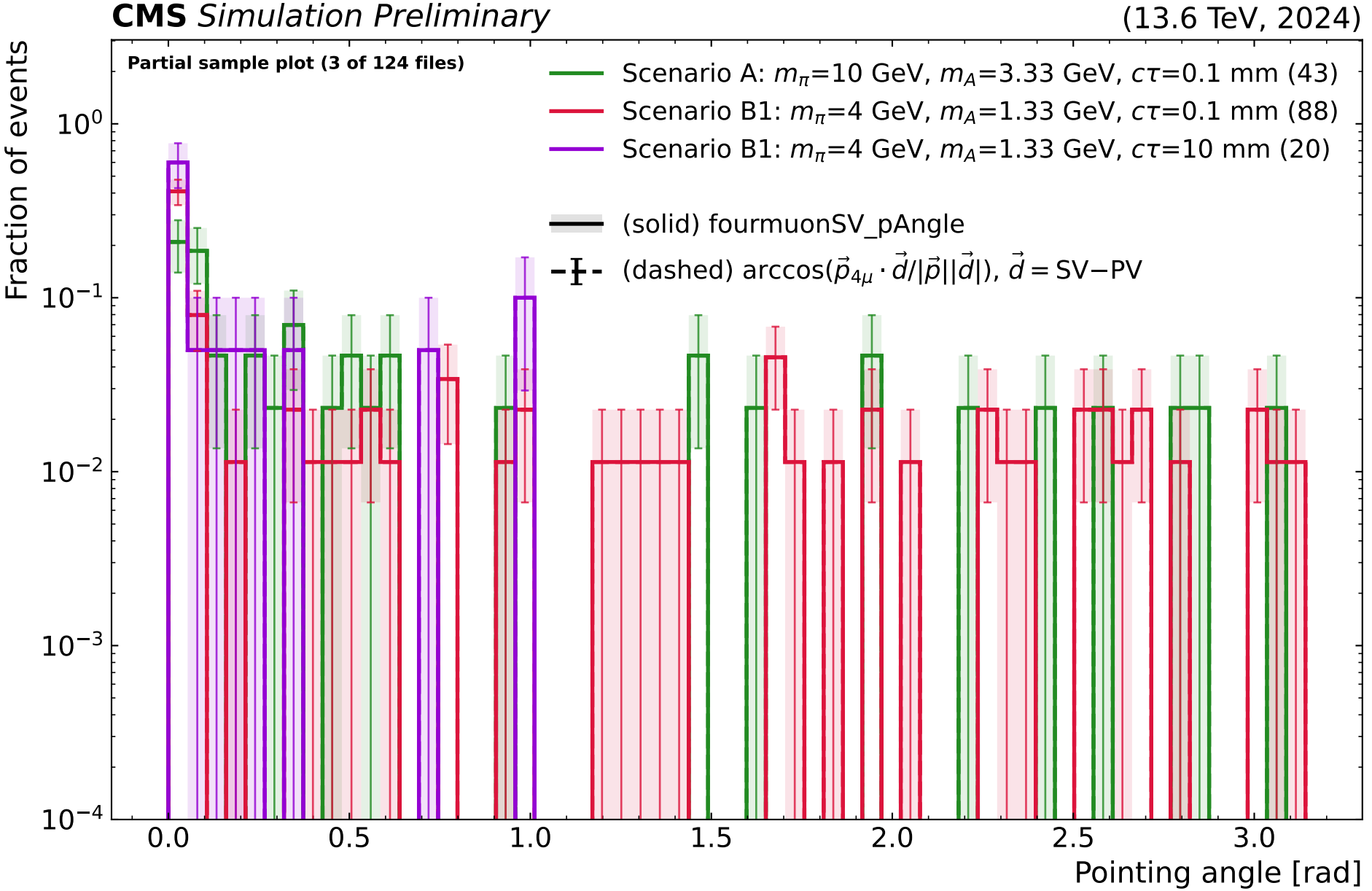
quadv: fourmuonSV + 2 consistent muonSV
trigger (Mu10 OR DoubleMu) + offline kinematic pre-selection
each distribution is normalised to unit area; bands/bars are statistical only



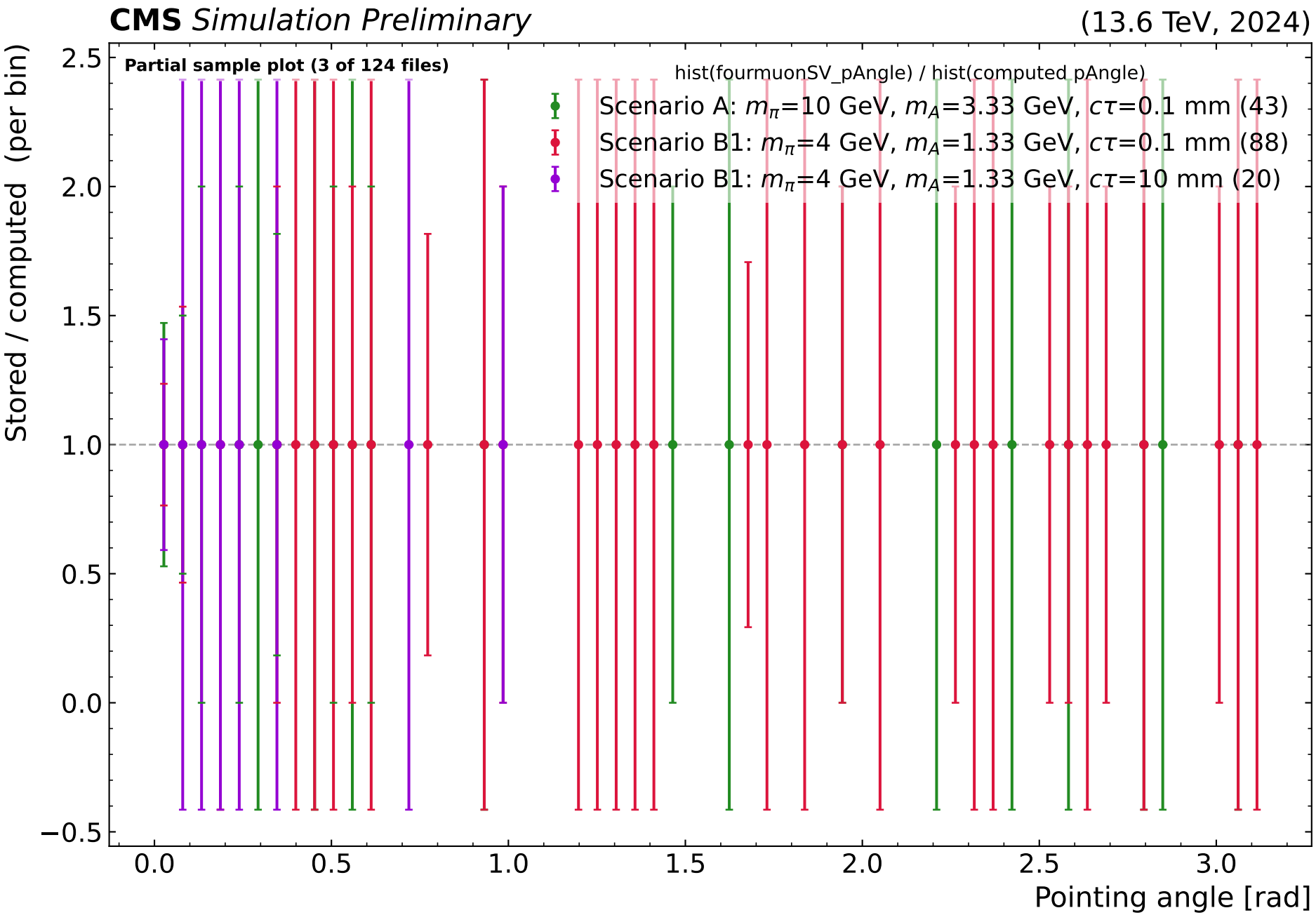
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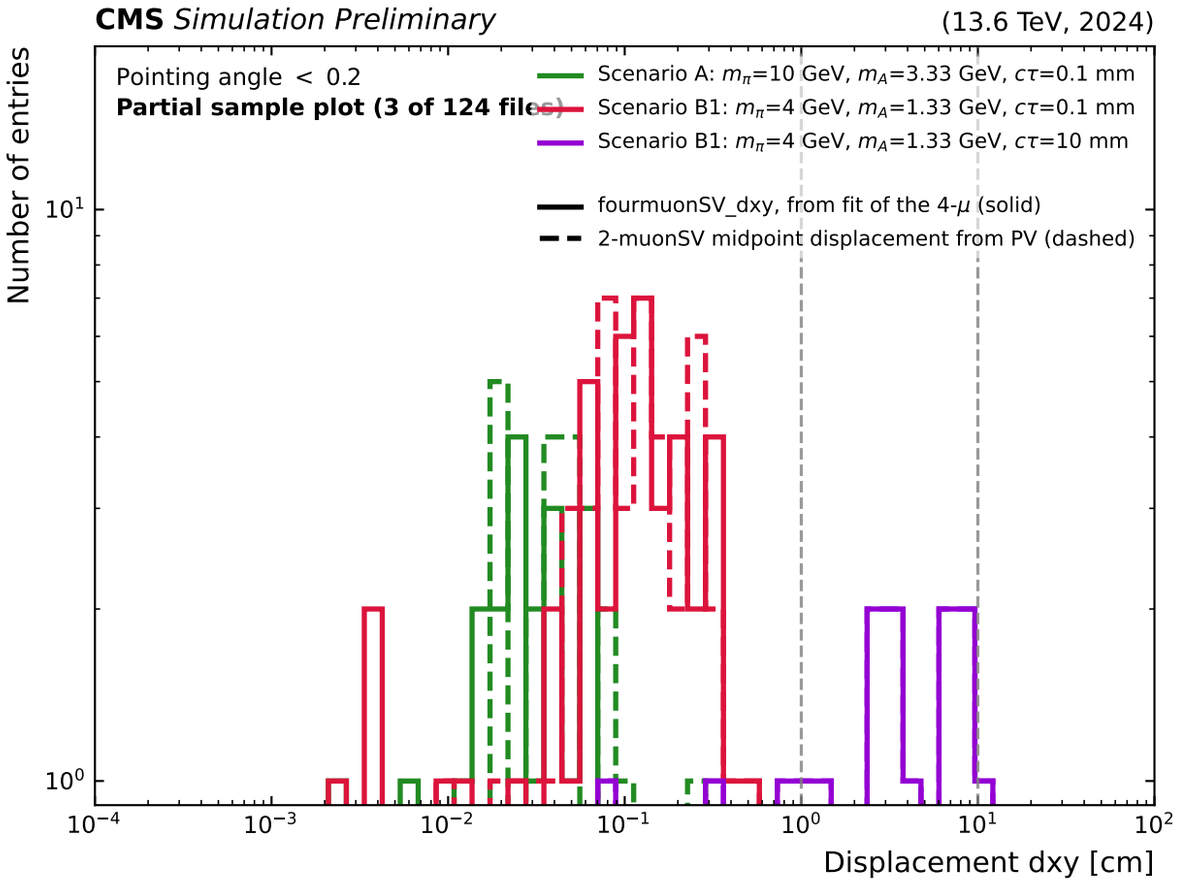


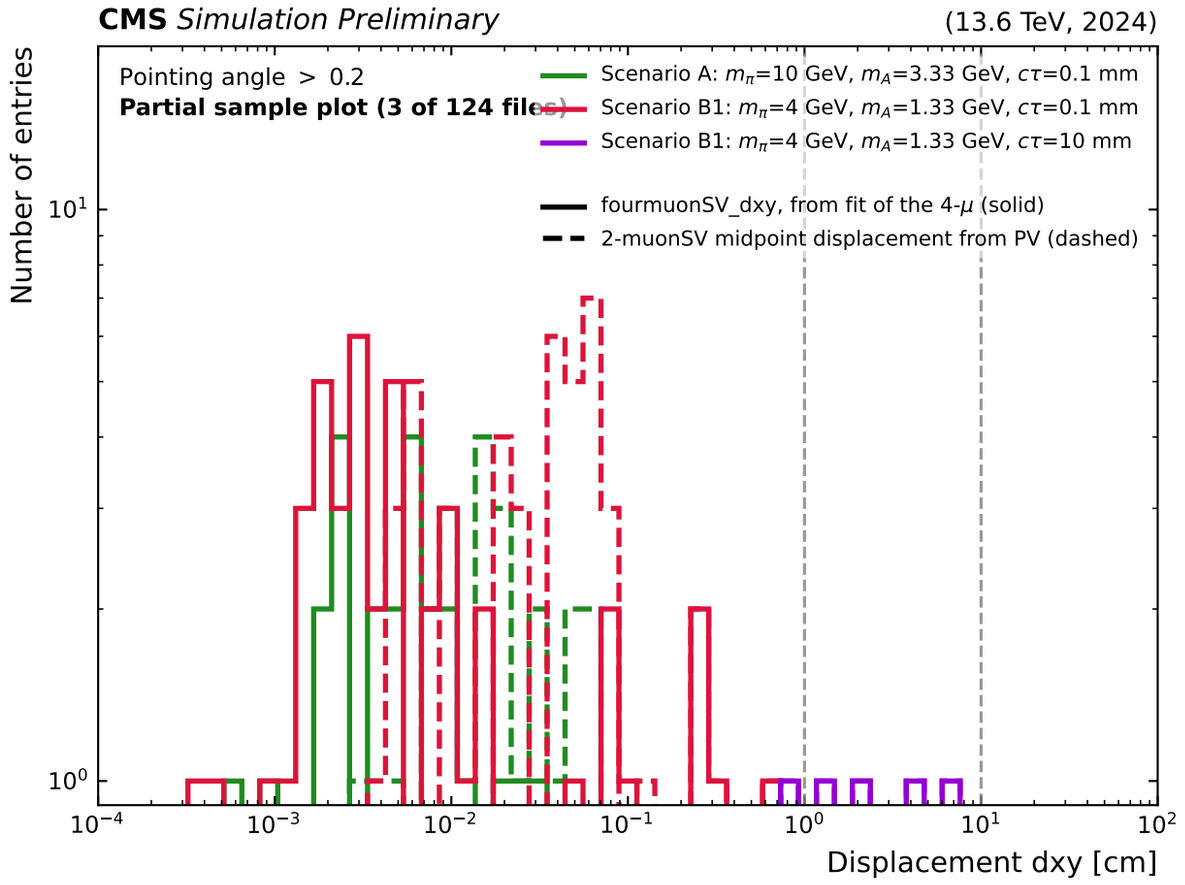
quadv: fourmuonSV + 2 consistent muonSV
trigger (Mu10 OR DoubleMu) + offline kinematic pre-selection



quadv category populations vs dxy, split by fourmuonSV_pAngle

solid: fourmuonSV_dxy, from fit of the 4-μ dashed: 2-muonSV midpoint displacement from PV grey lines: dxy category edges (1, 10 cm)



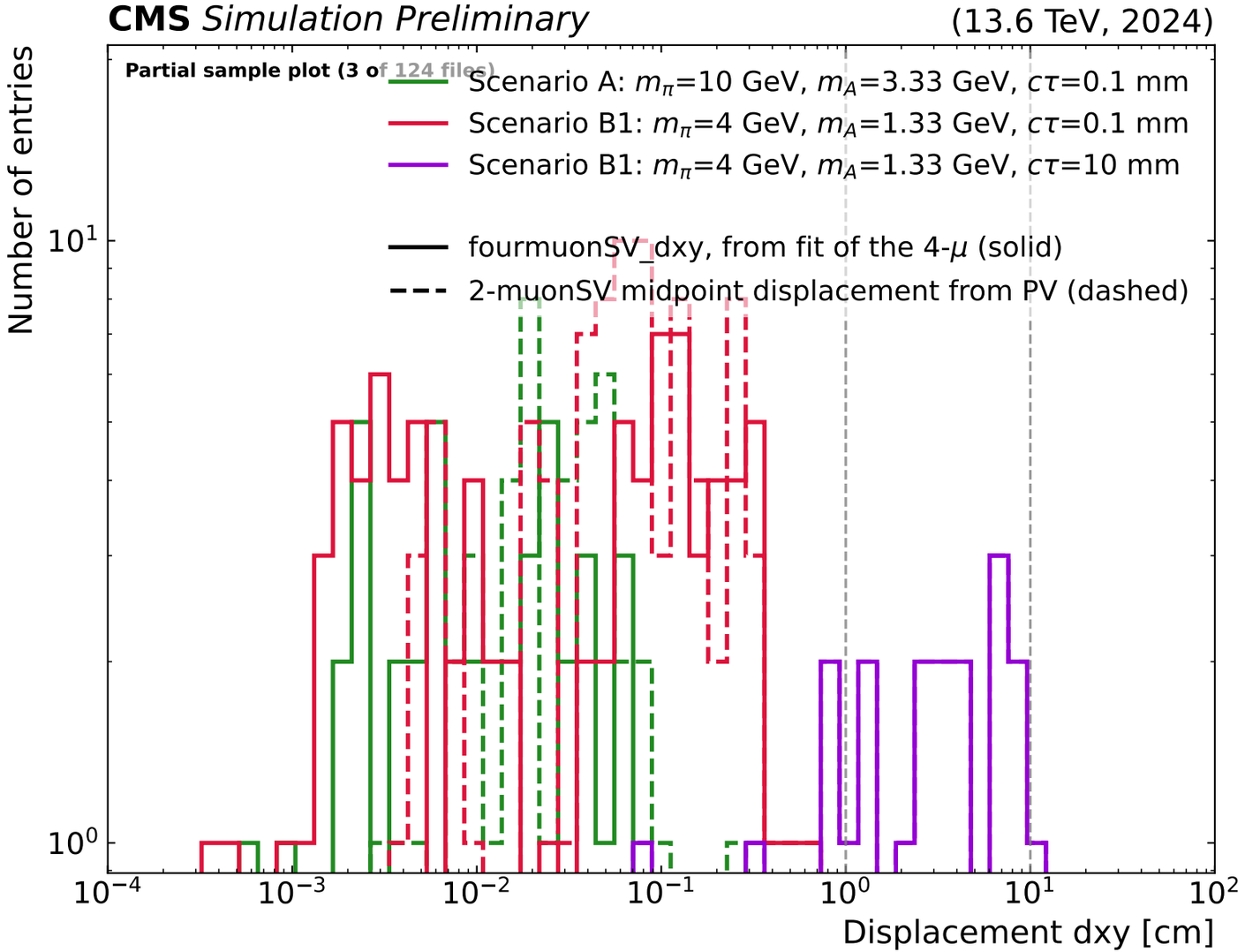


	dxy < 1	1--10	dxy > 10	Total
A $c\tau$=0.1 fourmuonSV	19	0	0	19
A $c\tau$=0.1 2muSV mid	19	0	0	19
B1 $c\tau$=0.1 fourmuonSV	44	0	0	44
B1 $c\tau$=0.1 2muSV mid	44	0	0	44
B1 $c\tau$=10 fourmuonSV	3	11	1	15
B1 $c\tau$=10 2muSV mid	3	11	1	15

	dxy < 1	1--10	dxy > 10	Total
A $c\tau$=0.1 fourmuonSV	24	0	0	24
A $c\tau$=0.1 2muSV mid	24	0	0	24
B1 $c\tau$=0.1 fourmuonSV	44	0	0	44
B1 $c\tau$=0.1 2muSV mid	44	0	0	44
B1 $c\tau$=10 fourmuonSV	1	4	0	5
B1 $c\tau$=10 2muSV mid	1	4	0	5

quadv category populations vs dxy (pAngle-inclusive)

solid: fourmuonSV_dxy, from fit of the 4-μ dashed: 2-muonSV midpoint displacement from PV grey lines: dxy category edges (1, 10 cm)



	dxy < 1	1--10	dxy > 10	Total
A cτ=0.1 fourmuonSV	43	0	0	43
A cτ=0.1 2muSV mid	43	0	0	43
B1 cτ=0.1 fourmuonSV	88	0	0	88
B1 cτ=0.1 2muSV mid	88	0	0	88
B1 cτ=10 fourmuonSV	4	15	1	20
B1 cτ=10 2muSV mid	4	15	1	20

Ratio of dxy distributions: fourmuonSV_dxy / (2-muonSV midpoint dxy)
grey dashed: dxy category edges (1, 10 cm); dotted: ratio = 1

